

Introduction to Artificial Intelligence

Lecture 5: Neural networks II

September 18, 2025



Terminology of AI

- What is machine learning?
- What is data science?
- What is the distinction between machine learning vs. data science?



Terminology of AI

- Machine learning vs. data science

Size of house (square feet)	# of bedrooms	Price (1000\$)
523	1	305
645	1	384
726	2	540
1088	3	653

A B

- ML: running an AI system (e.g., websites / mobile app)
- Homes with 3 bedrooms are more expensive than homes with 2 bedrooms of a similar size
 - Newly renovated homes have a 15% premium
 - New insights help you make better business decisions (DS)



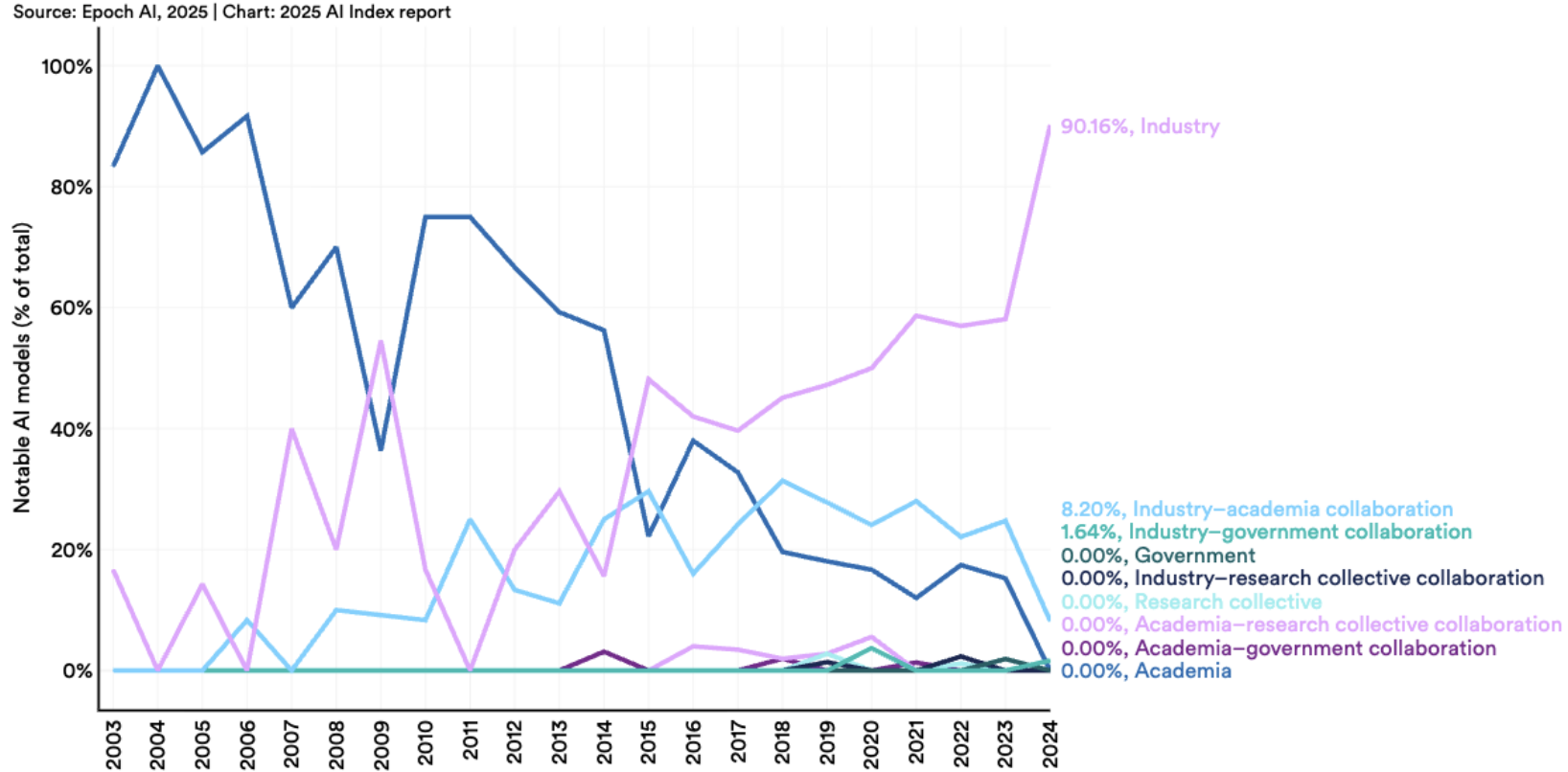
Terminology of AI

- Machine learning: “Field of study that gives computers the ability to learn without being explicitly programmed.” --- Arthur Samuel (1959)
- Data science: Science of extracting knowledge and insights from data



AI models

- Number of notable AI models by sector, 2003–24



AI models

- Number of notable AI models by organization, 2024

Source: Epoch AI, 2025 | Chart: 2025 AI Index report

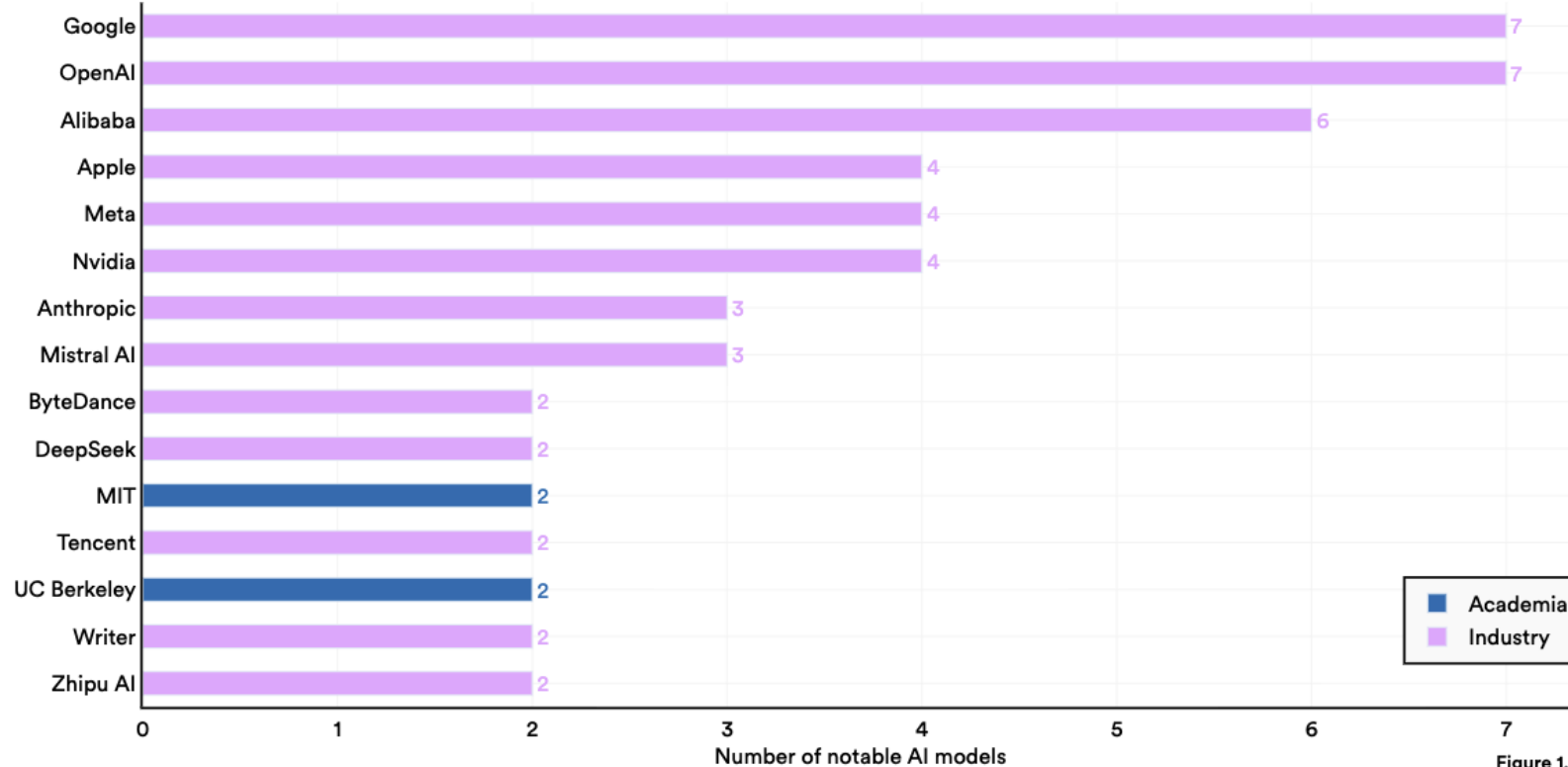


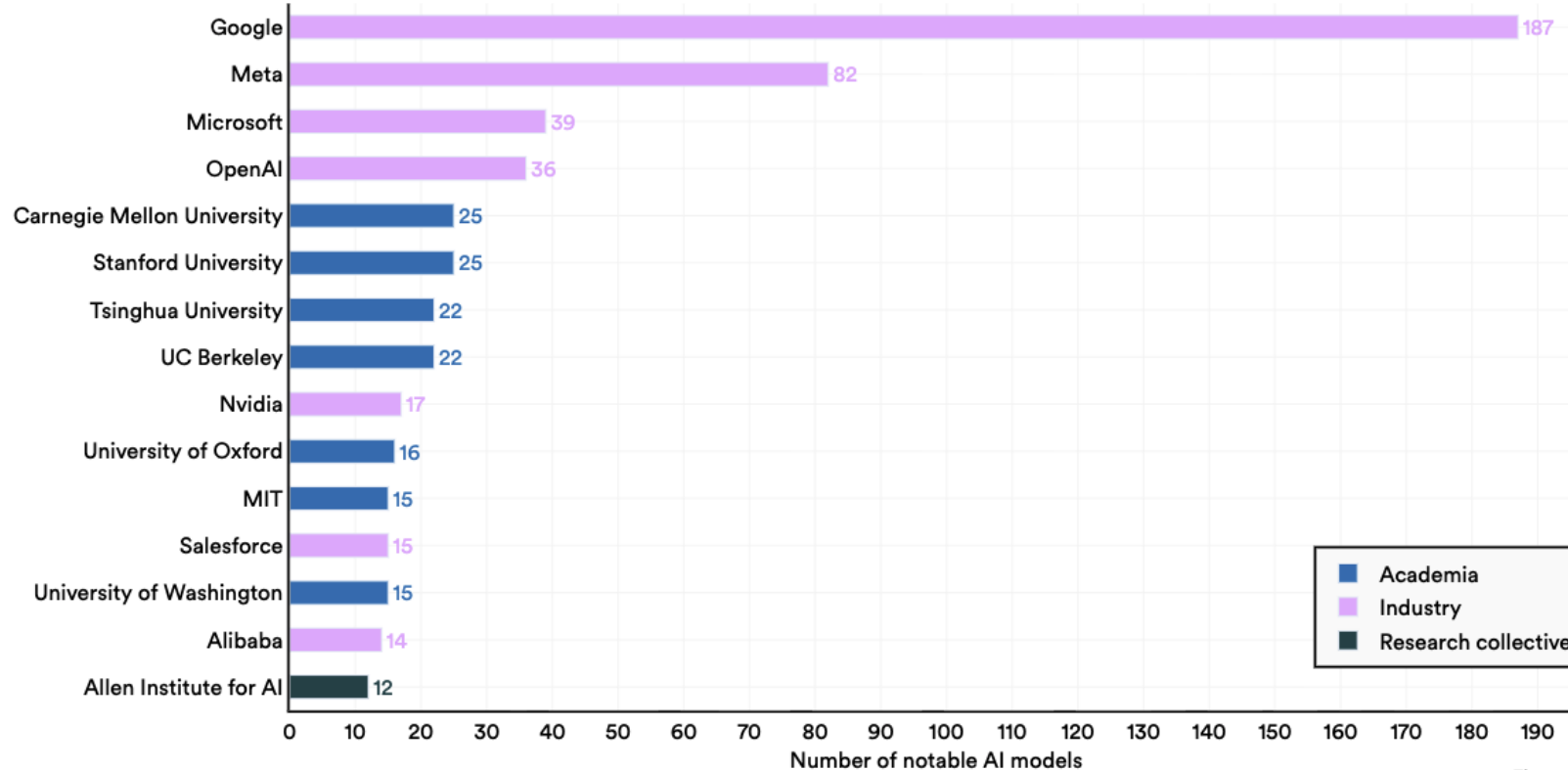
Figure 1.3.6¹⁹



AI models

- Number of notable AI models by organization, 2014–24 (sum)

Source: Epoch AI, 2025 | Chart: 2025 AI Index report



Model access via paid API subscriptions

Latest models

Contact sales

Start building

Opus 4.1

Most intelligent model for complex tasks

Input

\$15 / MTok

Output

\$75 / MTok

Prompt caching

Write

\$18.75 / MTok

Read

\$1.50 / MTok

Sonnet 4

Optimal balance of intelligence, cost, and speed

Input

Prompts ≤ 200K tokens

\$3 / MTok

Prompts > 200K tokens

\$6 / MTok

Output

Prompts ≤ 200K tokens

\$15 / MTok

Prompts > 200K tokens

\$22.50 / MTok

Prompt caching

≤ 200K tokens

Write

\$3.75 / MTok

Read

\$0.30 / MTok

> 200K tokens

Write

\$7.50 / MTok

Read

\$0.60 / MTok

Haiku 3.5

Fastest, most cost-effective model

Input

\$0.80 / MTok

Output

\$4 / MTok

Prompt Caching

Write

\$1 / MTok

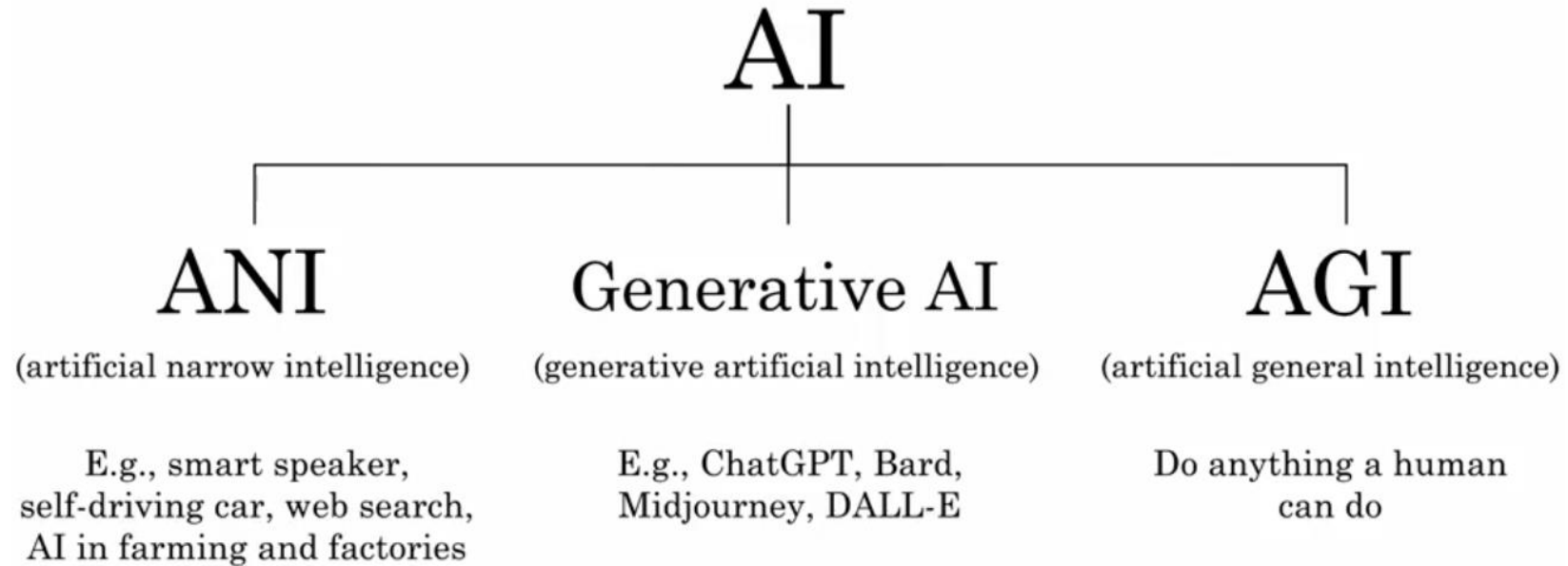
Read

\$0.08 / MTok

- Reference: <https://claude.com/pricing#api>



Demystifying AI



- “AI models” serve as the backbone technology behind all the above sectors



Recap: What are neural networks?

Input (A)	Output (B)	Application
Email	Spam? (0/1)	Spam filtering
Audio	Text transcripts	Speech recognition
English	Chinese	Machine translation
Ad, user info	Click? (0/1)	Online advertising
Image, radar info	Position of other cars	Self-driving cars
Image of phone	Defect? (0/1)	Visual inspection
Sequence of words	The next word	Chatbot



How large language models (LLMs) work

- LLMs are built by using supervised learning (A $\rightarrow$ B) to repeatedly predict the next word

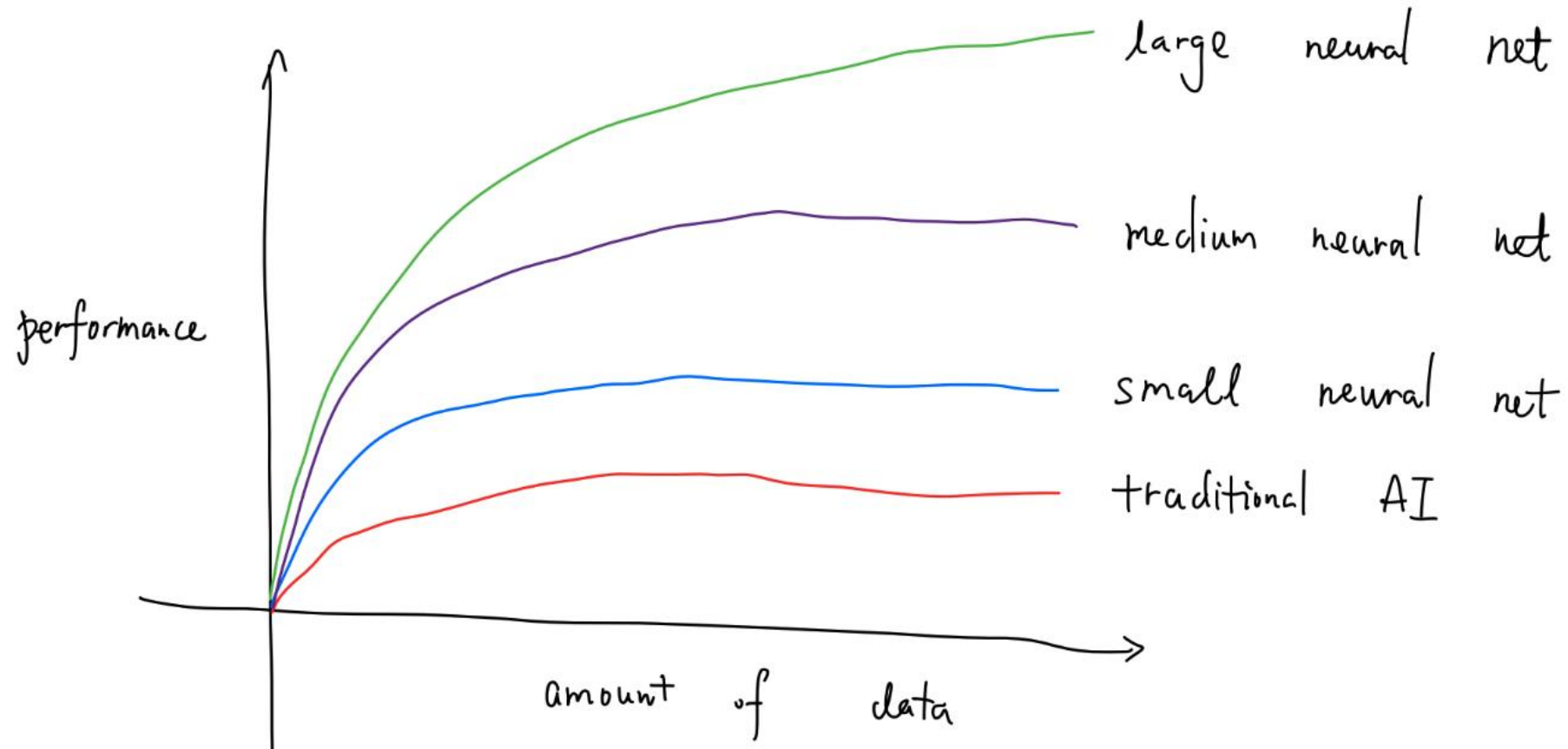
CS4100 is a class about artificial intelligence

Input (A)	Output (B)
CS4100	is
CS 4100 is	a
CS 4100 is a	class
CS 4100 is a class	about
CS 4100 is a class about	artificial
CS 4100 is a class about artificial	intelligence

- When we train a very large AI system on a lot of data (hundreds of billions of words), we get a large language model like ChatGPT



From neural networks to AI



- It helps to have more data, plus more compute



Example of a table of data (dataset)

Size of house (square feet)	# of bedrooms	Price (1000\$)
523	1	305
645	1	384
726	2	540
1088	2	653

A

B

- Regression models: map A to B
- Neural networks

image	label
	cat
	not cat
	cat
	not cat



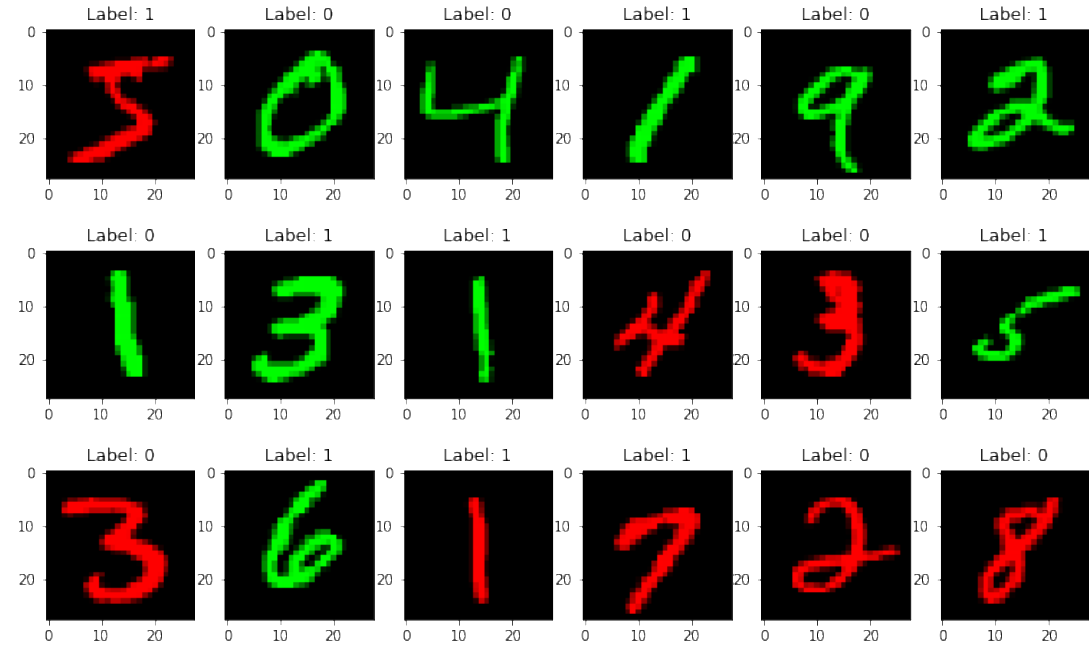
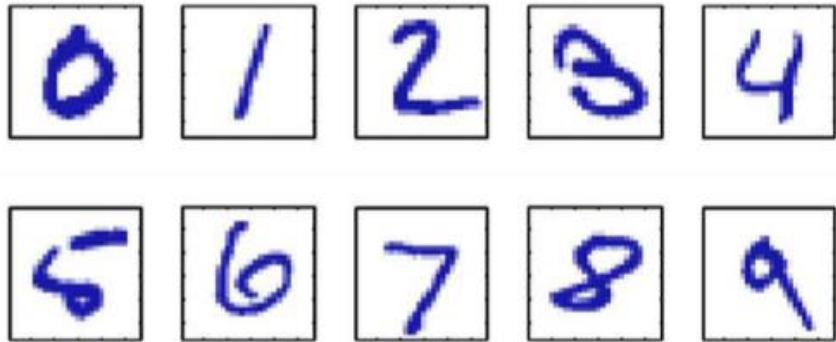
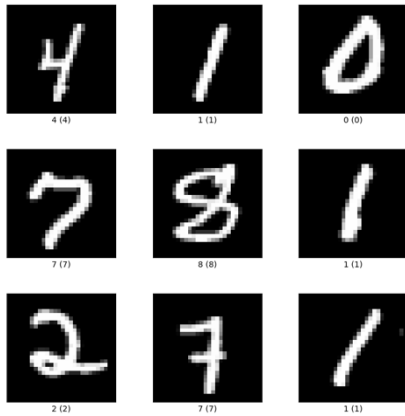
Lecture plan

- Introduction to convolutional neural networks
 - What is a convolutional layer?



Application I: Handwritten digit classification

- Classifying handwritten digits



Application II: Object recognition

- CIFAR-10

airplane



automobile



bird



cat



deer



dog



frog



horse



ship



truck



- ImageNet (1000 classes)



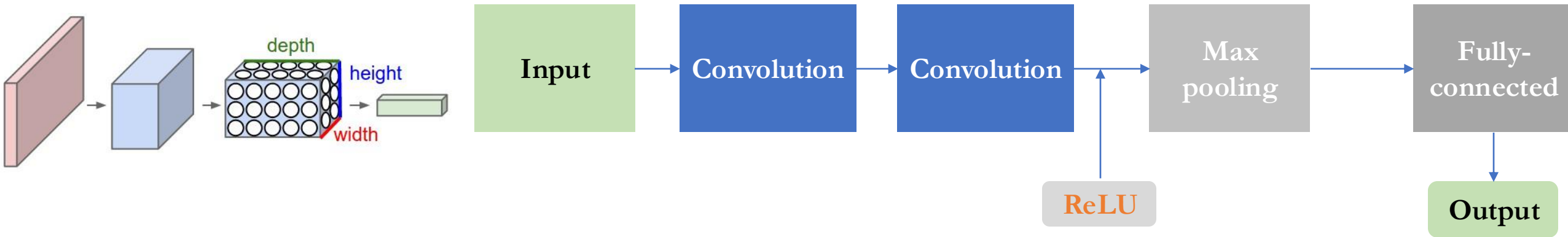
Why not using feedforward neural networks?

- Feedforward neural networks use fully-connected layers to transform the input
- **Fully-connected layers do not scale to large images**
 - A black-and-white digit in MNIST has size 28 by 28. A colored image in CIFAR-10 has size 32 by 32 by 3
 - For MNIST, a fully-connected neuron needs $28 \times 28 = 784$ weights
 - For CIFAR-10, a fully-connected neuron needs $32 \times 32 \times 3 = 3,072$ weights
 - Processing larger images requires more parameters



What is a convolutional neural network?

- In convolutional neural networks (CNN), a neuron only connects to a small local region of the image
 - Example: A colored (2D) image is specified by width, height, and depth

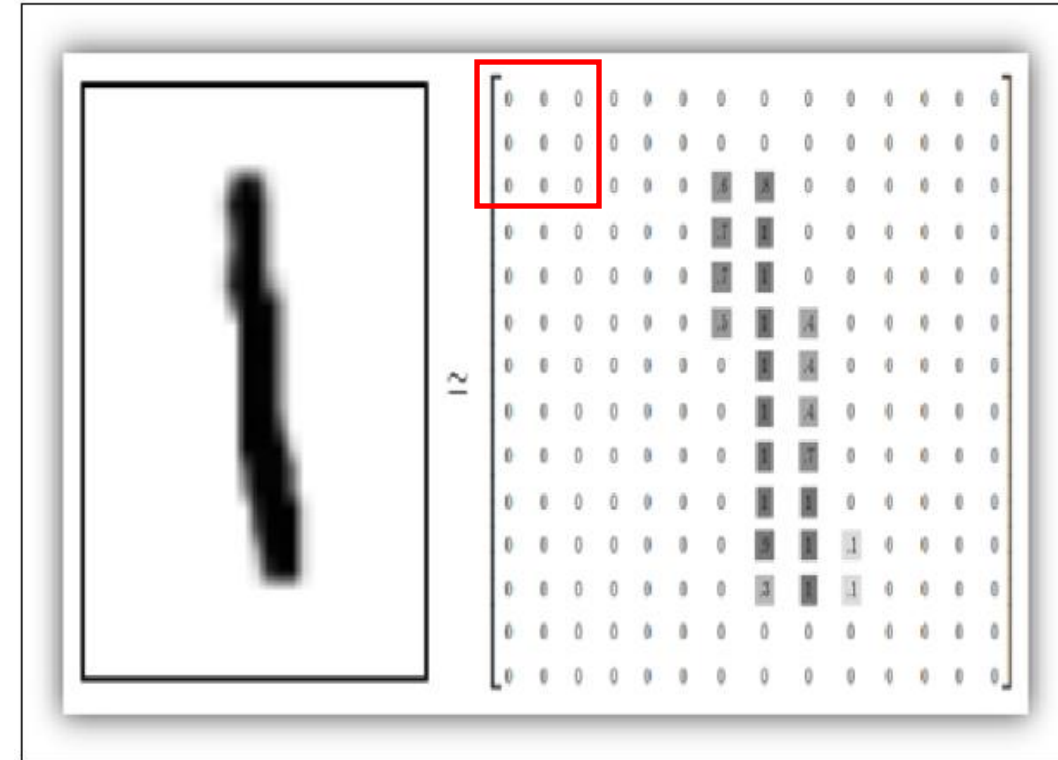
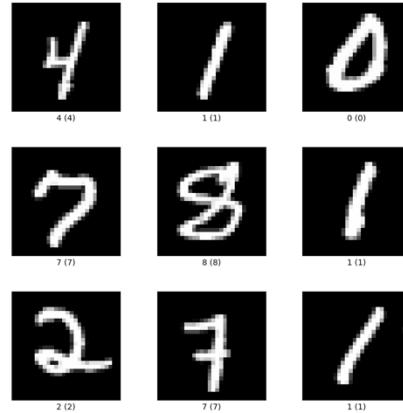


- A CNN involves a combination of the following types of layers
 - **Input layer:** Raw pixel values of the image
 - **Convolution layer:** Combine pixel values in a local region
 - **Pooling layer:** Down sample pixels
 - **Fully-connected layers:** Classification/prediction



Illustration of a convolution layer

- **Example** (MNIST)
 - Input size: 28 by 28
 - Convolutional layer:
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 0



- **First row, first patch**

Illustration of a convolution layer

- **Example** (MNIST)
 - **Input** size: 28 by 28
 - **Convolutional layer:**
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 0
 - **First row, second patch**

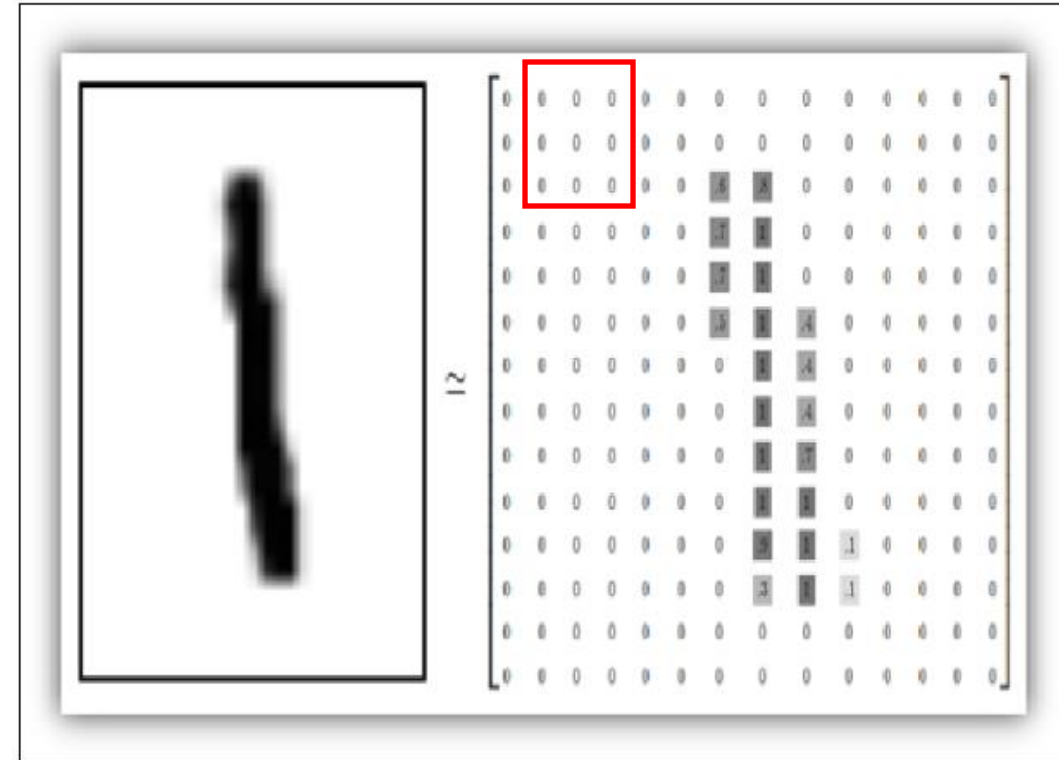


Illustration of a convolution layer

- **Example** (MNIST)
 - **Input** size: 28 by 28
 - **Convolutional layer:**
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 0
 - **First row, third patch**

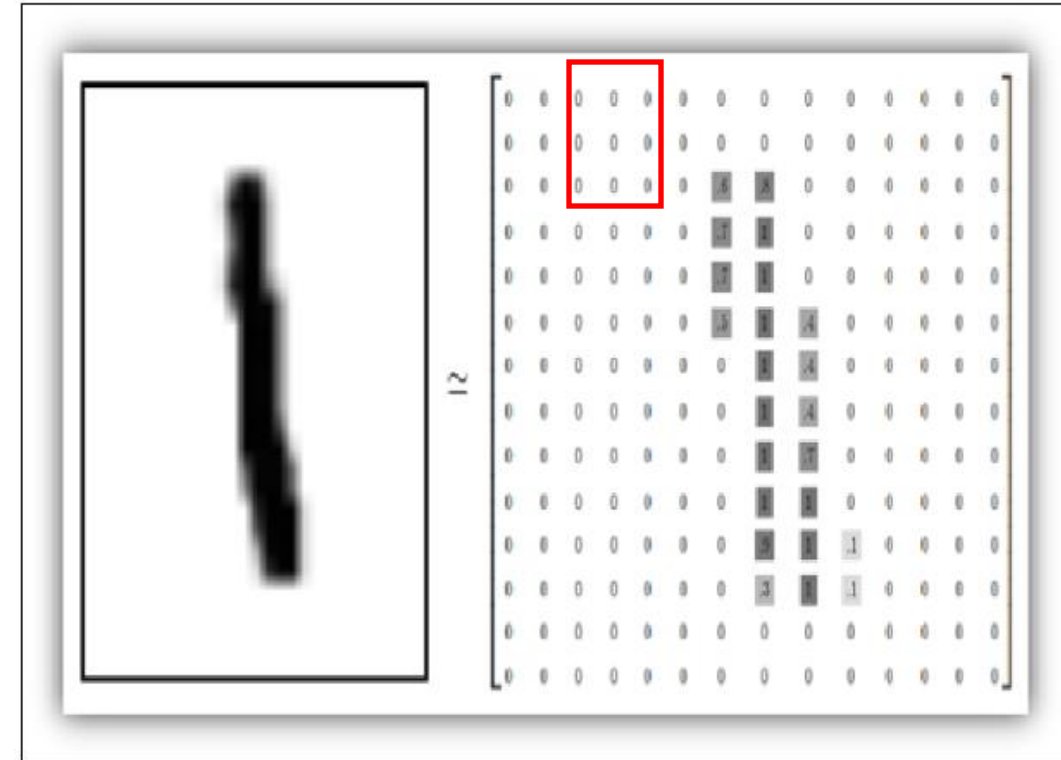


Illustration of a convolution layer

- **Example** (MNIST)
 - **Input** size: 28 by 28
 - **Convolutional layer:**
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 0
 - **First row, last patch**

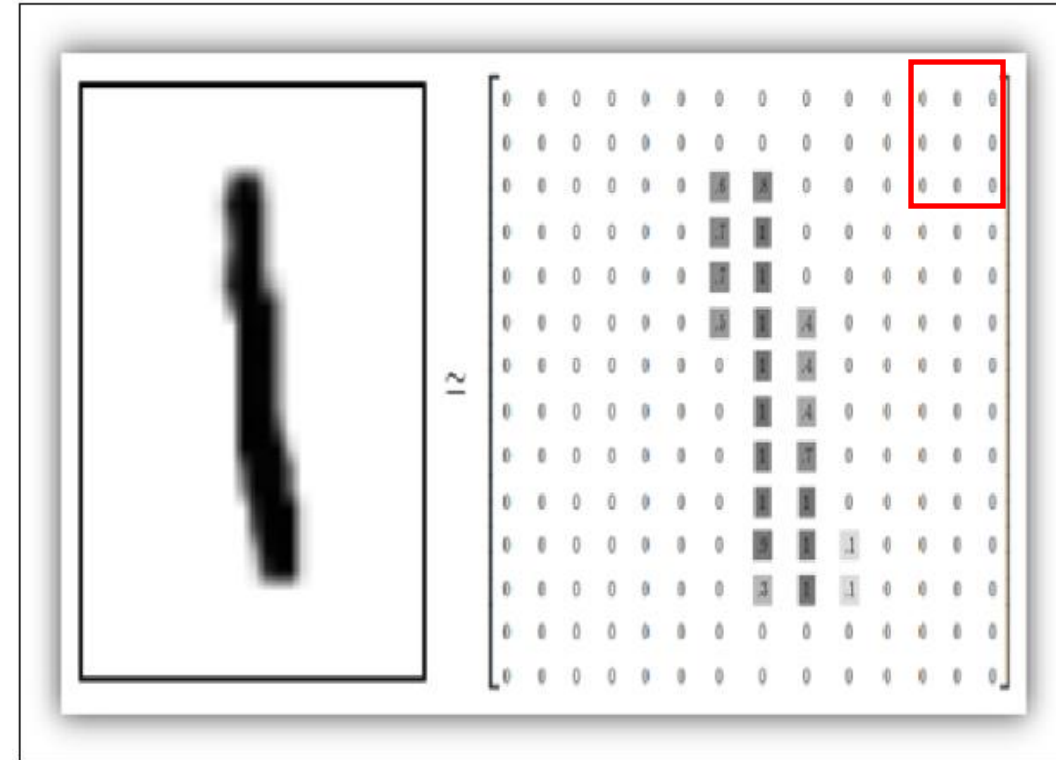


Illustration of a convolution layer

- **Example** (MNIST)
 - **Input** size: 28 by 28
 - **Convolutional layer:**
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 1
- **Second row, first patch**

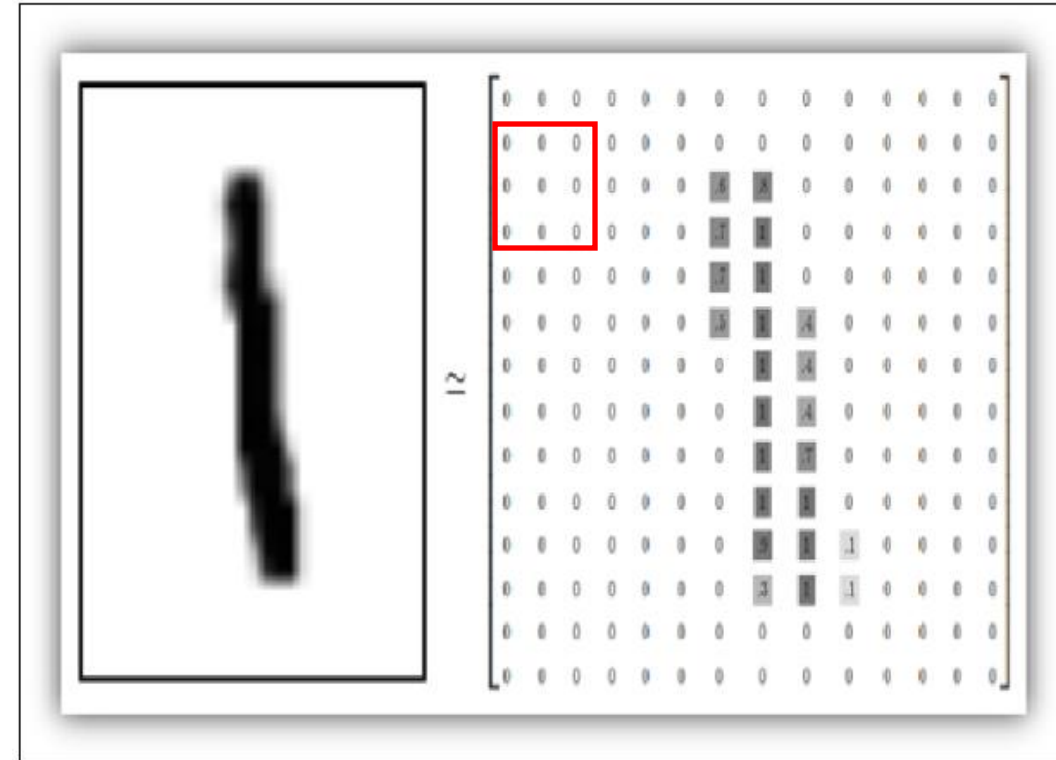


Illustration of a convolution layer

- **Example** (MNIST)
 - **Input** size: 28 by 28
 - **Convolutional layer:**
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 0
 - **Second row, second patch**

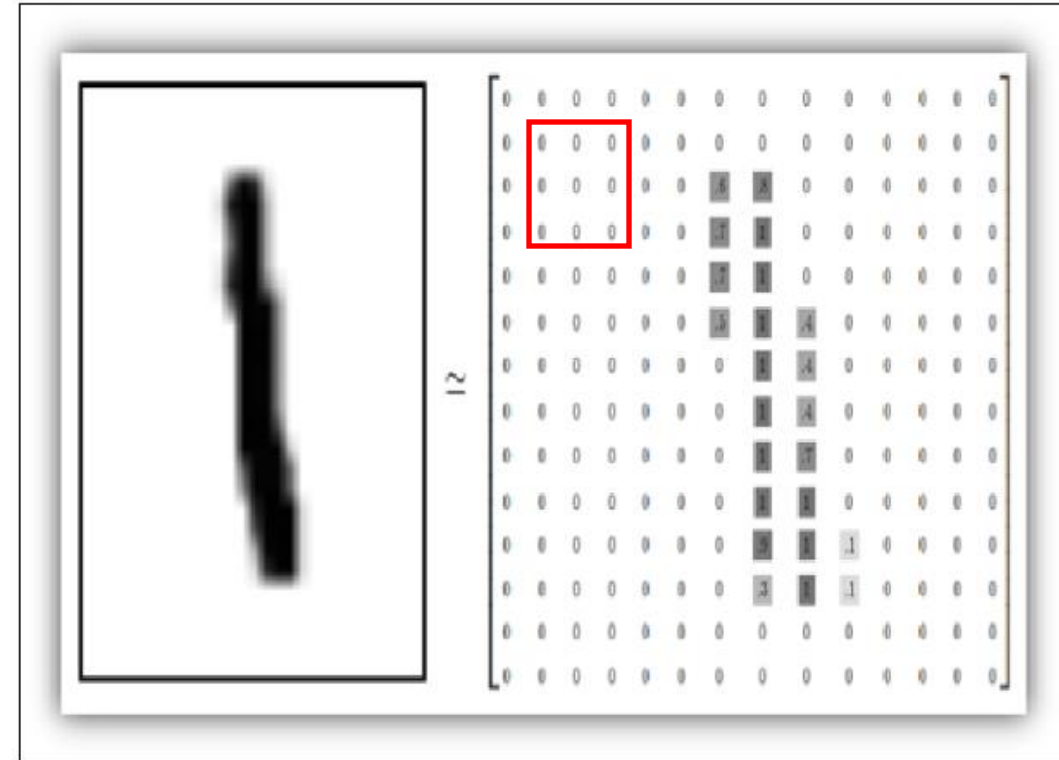
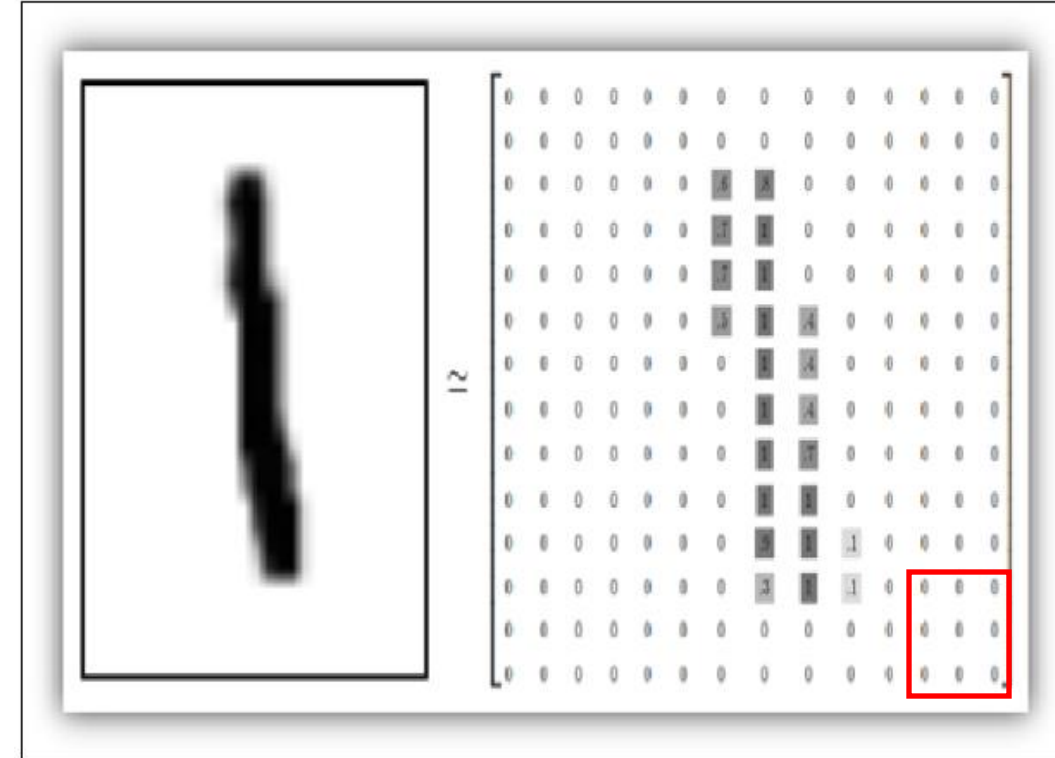
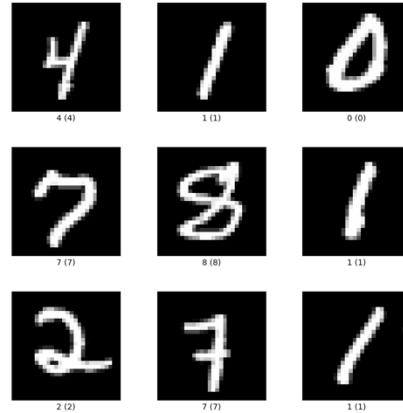


Illustration of a convolution layer

- **Example** (MNIST)
 - Input size: 28 by 28
 - **Convolutional layer:**
 - Filter size: (3, 3)
 - Stride: (1, 1)
 - Zero padding size: 0
 - **Last row, last patch**



- **Question:** What is the final output size?

Convolution layer

- **Filter (depth times width):** Larger filter captures coarser spatial patterns, while smaller filters capture finer spatial patterns
- **Stride (depth times width):** How often do we slide the filter? For example, when the stride is 1, we slide the filter one pixel at a time
- **Zero padding:** Pad the input with zeros around the border
- MNIST example: filter size (3, 3), stride size (1, 1), zero padding size 0



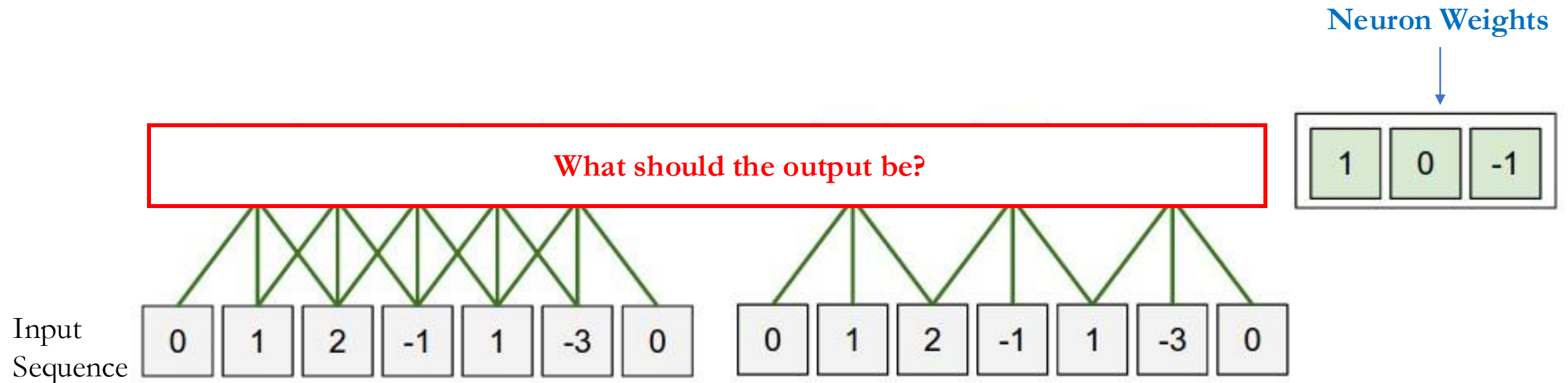
Lecture plan

- Introduction to convolutional neural networks
 - What is a convolutional layer?
 - **Illustrative examples**



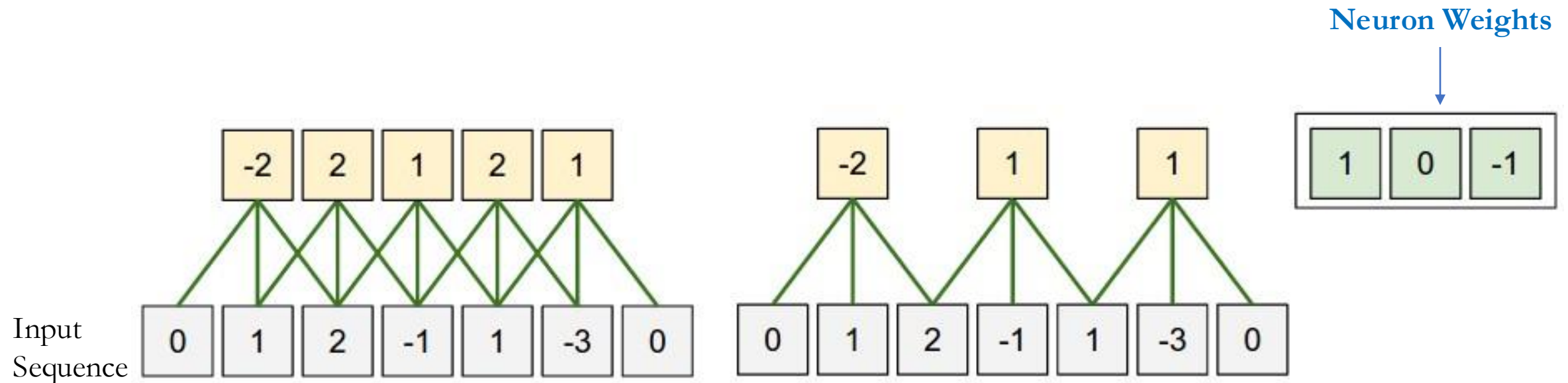
Illustration

- Input dimension is one, filter size is (3), stride is (1)
- Multiply the input with the neuron weights pixel-by-pixel



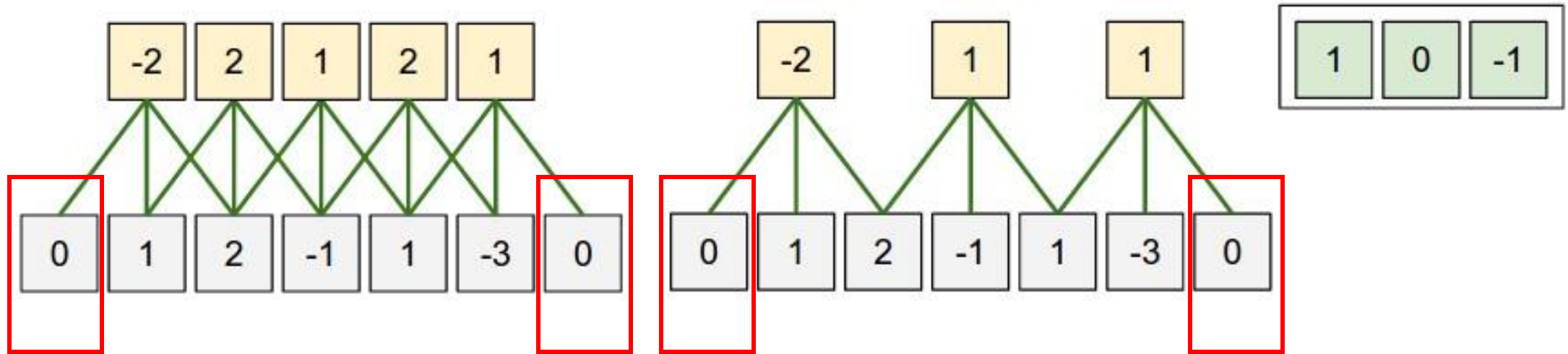
Illustration

- Illustration of spatial arrangement with a simplified example
 - Filter size is (3)
 - Stride is (1)



Explaining zero padding size

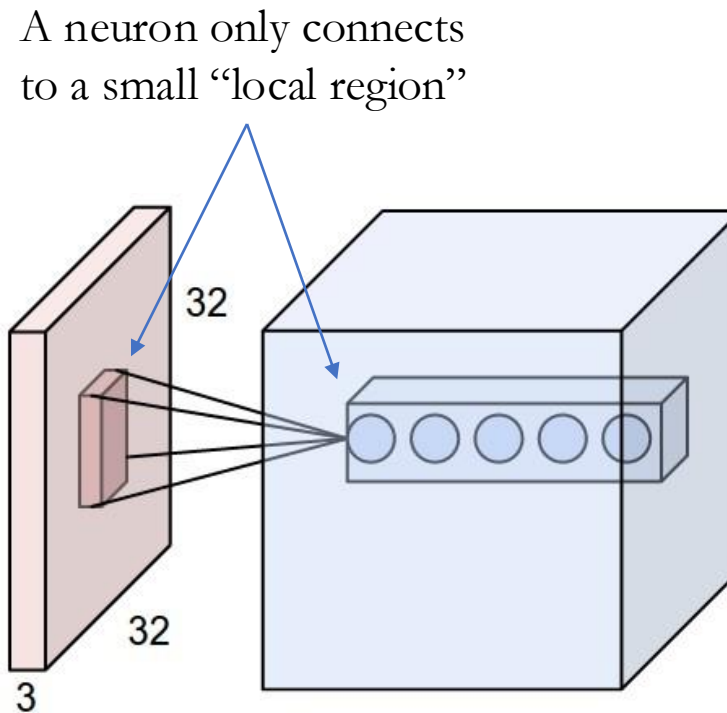
- This example uses a **single zero padding** on both left and right



- We can use zero padding to adjust the output dimension, e.g., in sentence classification, use zero padding for fixed (max) length sentences

Stride size

- Filter size and stride size must satisfy that: (image width – filter size) should be divisible by (stride size); Otherwise, add zero padding
- Illustrating the convolution operation for an image of size (32, 32, 3)



Summary

- Input: A 3D image of size (W_1, H_1, D_1)
- Convolution layer:
 - Number of filters K
 - Filter size F ($F \times F \times D_1$)
 - Stride size S
 - Zero padding size P
- Produces an output of size (W_2, H_2, D_2) . What is it?
 - $W_2 = \frac{W_1 - F + 2P}{S} + 1$
 - $H_2 = \frac{H_1 - F + 2P}{S} + 1$
 - $D_2 = K$
- With parameter sharing, $F \times F \times D_1$ weights per filter, for a total of $(F^2 \times D_1) \times K$ weights



Lecture plan

- Introduction to convolutional neural networks
 - What is a convolutional layer?
 - Illustrative examples
 - **Numpy examples**



Numpy example

- **Input: numpy array X**
 - $X.shape = (11,11,4)$
- **Convolution layer**
 - Number of filters: $K = 2$
 - Filter size: $5 \times 5 \times 4$
 - Stride size: 2×2
 - Zero padding size: 0
- **Output:** Denote as V
 - Output width and height: $\frac{11-5}{2} + 1 = 4$
 - Depth: 2



Numpy example

- First depth slice, along the first column: Filter parameters W_0 , Bias b_0 .
 $W_0.shape = (5, 5, 4)$
 - $V[0,0,0] = np.sum(X[:5, :5, :] * W_0) + b_0$
 - $V[1,0,0] = np.sum(X[5:10, :5, :] * W_0) + b_0$
 - $V[2,0,0] = np.sum(X[10:15, :5, :] * W_0) + b_0$
 - $V[3,0,0] = np.sum(X[15:20, :5, :] * W_0) + b_0$



Numpy example

- For a different neuron: Filter parameters W_1 , bias b_1
 - $V[0,0,1] = np.sum(X[:5, :5, :] * W_1) + b_1$
 - $V[1,0,1] = np.sum(X[2:7, :5, :] * W_1) + b_1$
 - $V[2,0,1] = np.sum(X[4:9, :5, :] * W_1) + b_1$
 - $V[3,0,1] = np.sum(X[6:11, :5, :] * W_1) + b_1$
 - Question: how do we calculate $V[0,1,1]$ and $V[2,3,1]$?



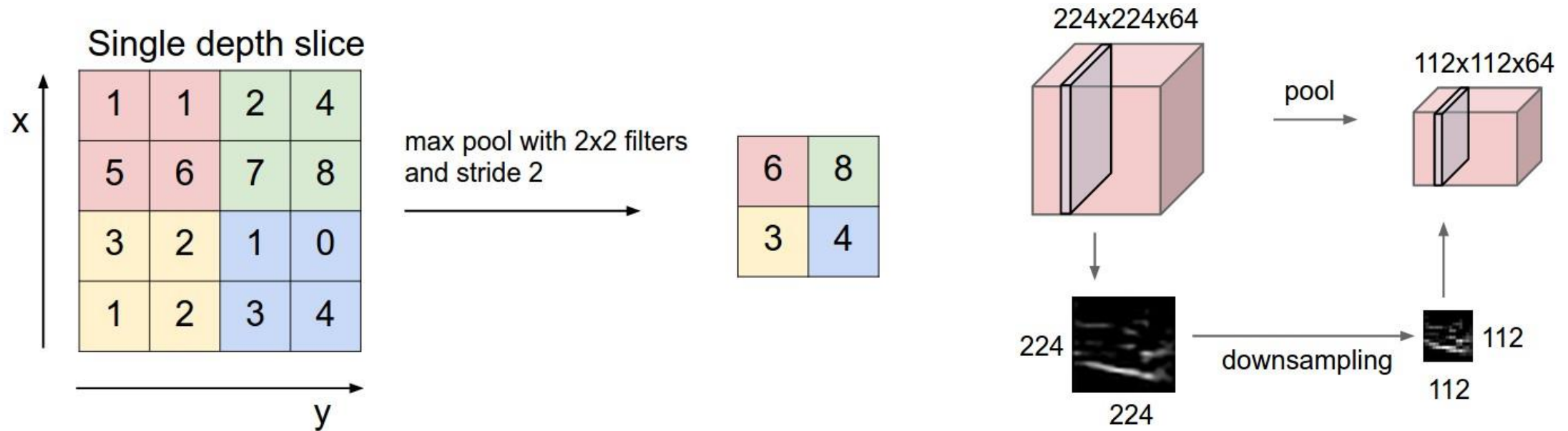
Lecture plan

- Introduction to convolutional neural networks
 - **Pooling and downsampling**



Pooling layer

- **Pooling** reduces the spatial size of the input: Insert a pooling layer between convolution layers

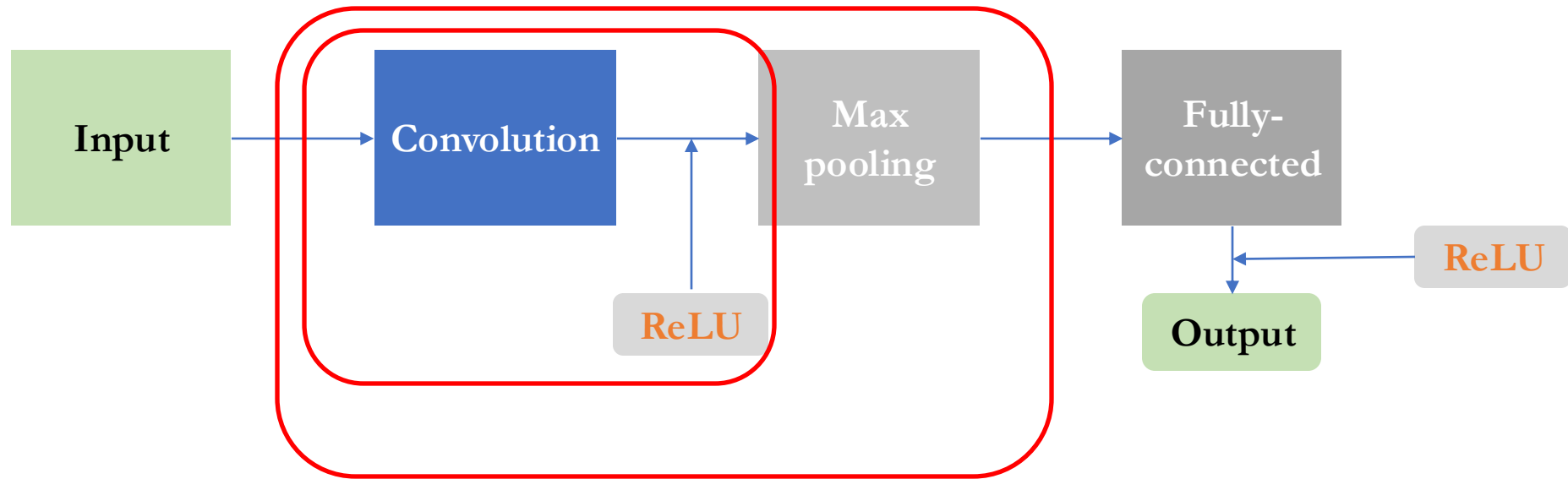


Pooling layer

- Input: An image of size (W_1, H_1, D_1)
- Pooling layer
 - Filter size F
 - Stride size S
- Output size: (W_2, H_2, D_2)
 - $W_2 = \frac{W_1 - F}{S} + 1$
 - $H_2 = \frac{H_1 - F}{S} + 1$
 - $D_2 = D_1$
- Previous example: $F = 2$ and $S = 2$



Summary



- A deep CNN architecture involves multiple convolution and pooling layers



Residual networks

- Residual networks are a popular and successful approach to building very deep networks

- Main idea: Let's say we map the information from layer $i - 1$ to layer i

$$z^{(i)} = f(z^{(i)}) = g(W^{(i)} z^{(i-1)})$$

- The key idea of residual networks is that a layer should perturb the representation from the previous layer rather than replace it entirely

$$z^{(i)} = g(z^{(i-1)} + f(z^{(i)}))$$



Applications

- Convolutional neural networks are most often used for computer vision
 - The AlexNet deep learning system in the 2012 ImageNet competition revolutionized the field
 - The ImageNet competition was a supervised learning task with 1,200,000 images in 1,000 different categories, and systems were evaluated on the “top-5” score—how often the correct category appears in the top five predictions
- Convolutional neural networks are also useful for text classification
 - Given a sentence, predict a certain label based on extracting meaning from the sentence

